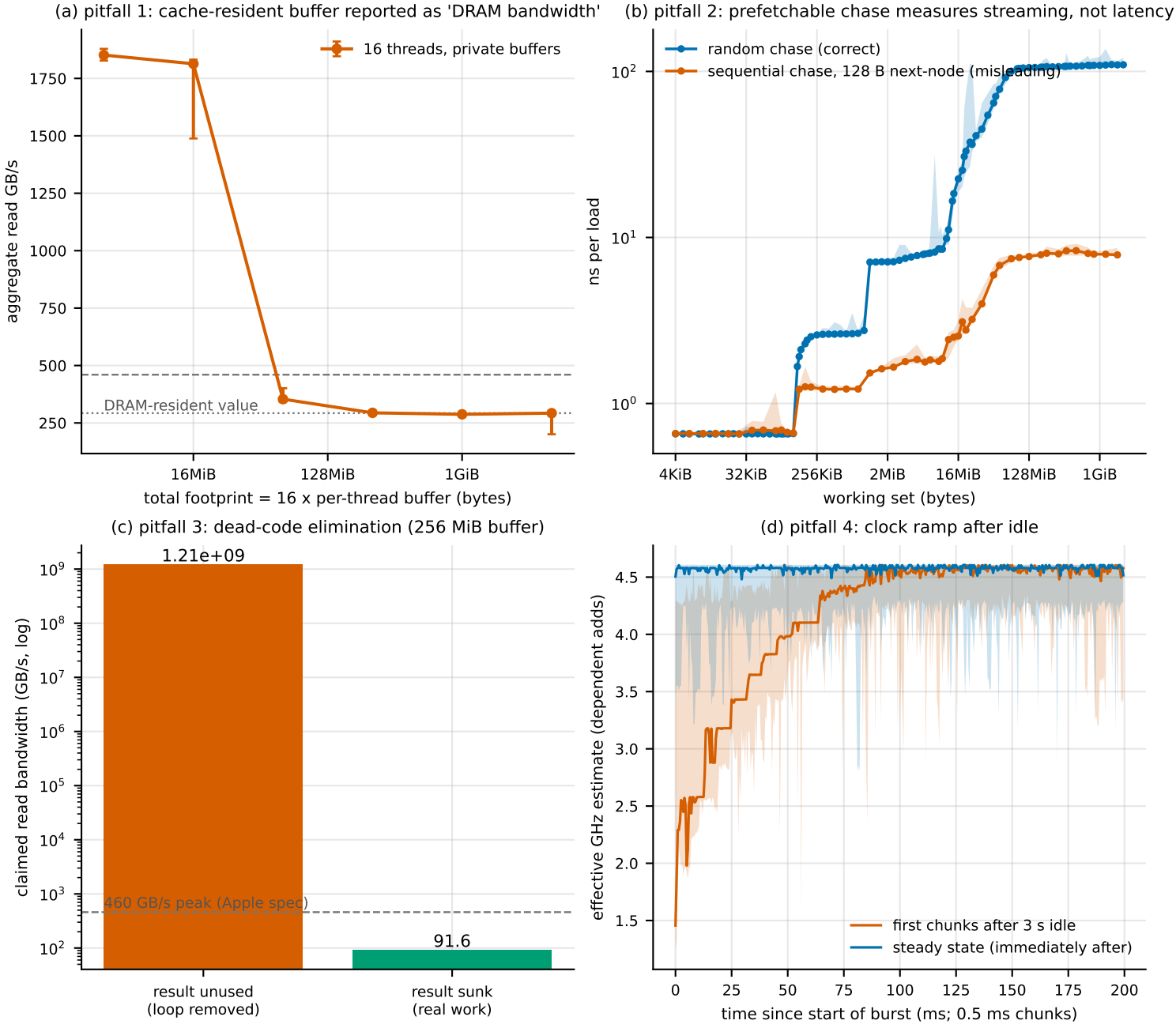
(a) pitfall 1: cache-resident buffer reported as 'DRAM bandwidth'
16 threads, private buffers
aggregate read GB/s
1750
1500
1250
1000
750
500
250
DRAM-resident value
16MiB
128MiB
1GiB
total footprint = 16 x per-thread buffer (bytes)
(b) pitfall 2: prefetchable chase measures streaming, not latency
random chase (correct)
sequential chase, 128 B next-node (misleading)
ns per load
10^2
10^1
10^0
4KiB
32KiB
256KiB
2MiB
16MiB
128MiB
1GiB
working set (bytes)
(c) pitfall 3: dead-code elimination (256 MiB buffer)
1.21e+09
claimed read bandwidth (GB/s, log)
10^9
10^8
10^7
10^6
10^5
10^4
10^3
10^2
460 GB/s peak (Apple spec)
91.6
result unused
(loop removed)
result sunk
(real work)
(d) pitfall 4: clock ramp after idle
effective GHz estimate (dependent adds)
4.5
4.0
3.5
3.0
2.5
2.0
1.5
first chunks after 3 s idle
steady state (immediately after)
0
25
50
75
100
125
150
175
200
time since start of burst (ms; 0.5 ms chunks)